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ACTION CONSTRAINTS EFFECTS ON DISTANCE PERCEPTION: META-ANALYSIS

BACKGROUND

According to Action Constraint Theories (e.g., Witt, 2011; Morgado & Palluel-Germain, 2015), people perceive distances according to the variables defining their action abilities. For instance, the easier people can reach an object, the closer they perceive it. Although many studies supported these theories, the question of the existence and magnitude of such action constraint effect on perceived distance is still highly debated (Firestone & Scholl, 2016). Additionally, some authors argued that action constraint effects reflect experimental demand bias on judgement rather than genuine perceptual effects (e.g., Durgin, Klein, Spiegel, Strawser & Williams, 2012).

OBJECTIVES

(1) Estimation of the **overall effect size** of action constraints on distance perception

(2) Estimation of two moderators (i.e., type of measure and experimental design) that could indicate that **experimental demand** plays a role in these effects

METHOD

INCLUSION CRITERIA

(1) Dependent variable = measure of distance estimation
(2) Independent variable = manipulation of an action constraint
(3) Distance estimation from an egocentric point of view
(4) Others ability manipulation excluded

COMPUTATION OF EFFECT SIZES

$$Cohen's\ ds = \frac{meanHC - meanLC}{\sqrt{(nHC - 1 * sdHC^2 + nLC - 1 * sdLC^2)/(nHC + nLC - 2)}}$$

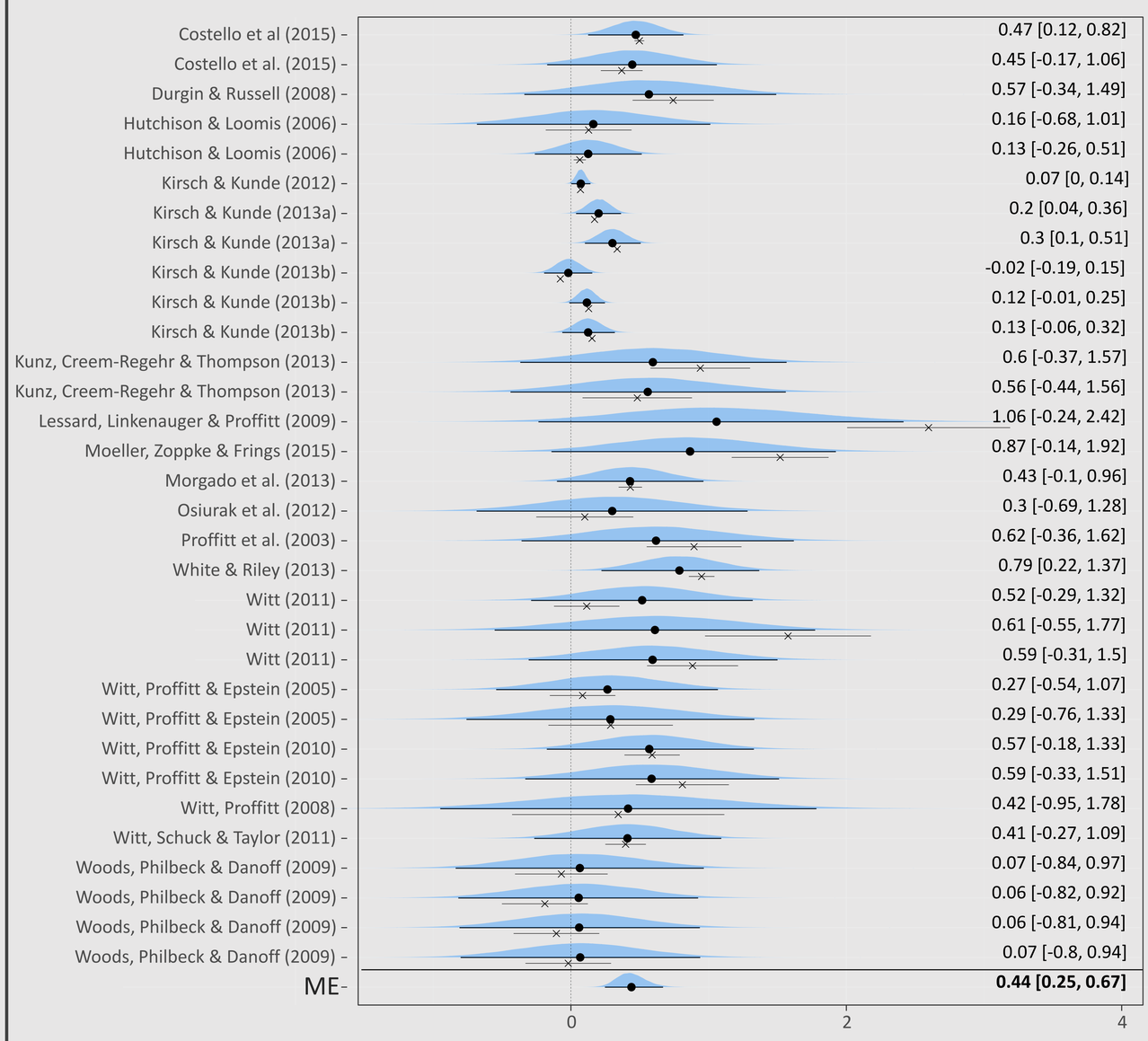
$$Cohen's\ drm = \frac{meanHC - meanLC}{\sqrt{sdHC^2 + sdLC^2 - 2 * r * sdHC * sdLC)/2 * 1 - r}}$$

META-ANALYTIC MODEL

Three-levels meta-analytic Bayesian model to estimate the overall effect size, and standard deviation between papers (τ_p) and between studies (τ_e)

HC = High Action Constraints, LC = Low Action Constraints

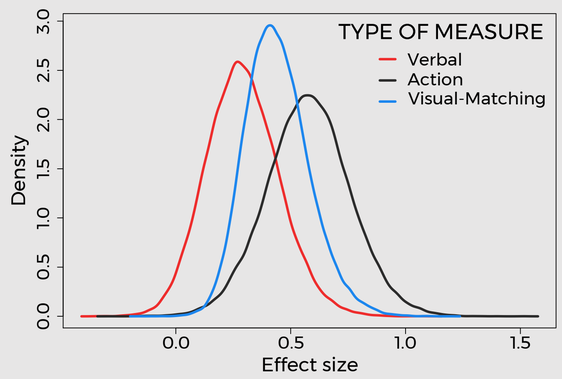
OVERALL EFFECT

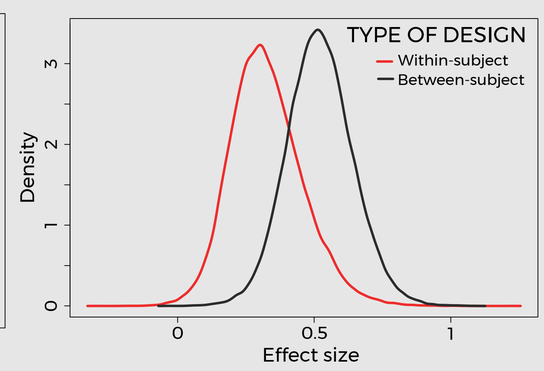


32 outcomes extracted from 19 articles (N = 770), g = 0.44 [0.25, 0.67], τ_p = 0.32, 95% CrI [0.06, 0.58], τ_e = 0.11, 95% CrI [0.01, 0.30], BF_{10} = 389. The effect size of action constraint on distance perception is medium (g = 0.44) and the Bayes Factor suggest that the existence of the effect is 389 more likely than its non-existence.

MODERATOR ANALYSES

POSTERIOR DISTRIBUTION OF THE EFFECT SIZE

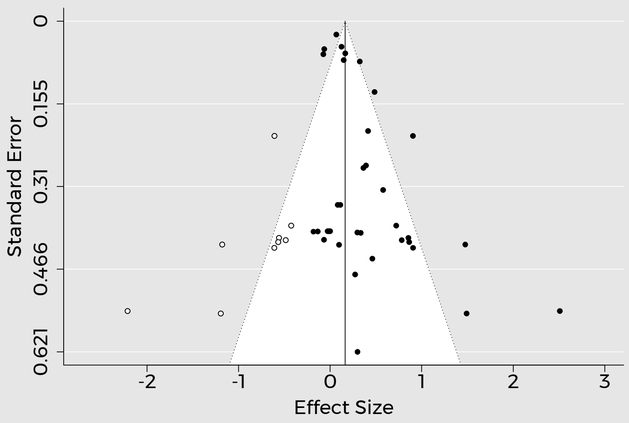




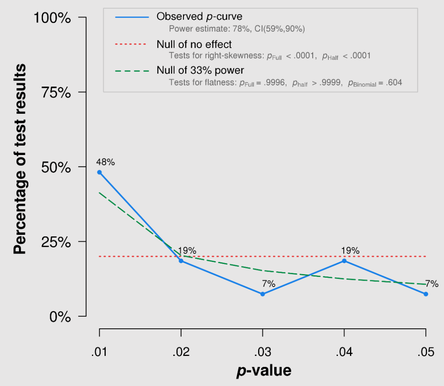
Verbal measures ($N_{expe.verbal}$ = 11) do not produce greater effect sizes than other measures ($N_{expe.action}$ = 8; $N_{expe.visual\text{-}matching}$ = 16) , and within-subject design (N_{expe} = 10) do not produce greater effect sizes than between-subject design (N_{expe} = 22).

ADDITIONAL ANALYSES

FUNNEL PLOT



P-CURVE



The trim-and-fill method indicated that 8 unpublished studies are missing on the left side of the funnel plot. Including these studies would reduce the effect size to g = 0.19 [-0.02, 0.38].
The observed distribution of p-values (p-curve) from our database is close to a distribution that would be expected with a true effect size.

CONCLUSIONS

Our meta-analysis indicates that people estimate a target farther away of 0.44 standard deviation unit when they are in high constraint condition than in low constraint condition. However, even if our estimation seems to be inflated by publication bias, our moderator analysis does not support the experimental demands hypothesis. **This meta-analysis supports the action constraint theories according to which action constraints influence perceived distance.**

REFERENCES

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